Quick & Dirty Data Management

The 5 Things You Need to be Doing Now!

Data Management Services
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Intended learning outcomes

- At the end of this session, attendees should be able to identify and describe specific actions they can take now to enhance their data management practices.
- Attendees will be able to compare their current data practices with good data practices and report on discrepancies and identify resources to support practice change.
- Attendees will find data management more approachable as a practice, and less overwhelming.

For more information on what are Intended Learning Outcomes and why we include them, see https://tll.mit.edu/help/intended-learning-outcomes.

Data Management Services @ MIT Libraries

Workshops

- Upcoming Classes & Events: https://libraries.mit.edu/news/events/

Web guide

- http://libraries.mit.edu/data-management

Consultations (individual or research groups)

- includes help with creating data management plans, setting up file organization, finding data repositories, etc.

Contact: data-management@mit.edu

The 5 things you can do now

1. Backups
2. File naming and organization
3. Documenting
4. Security
5. Future proofing

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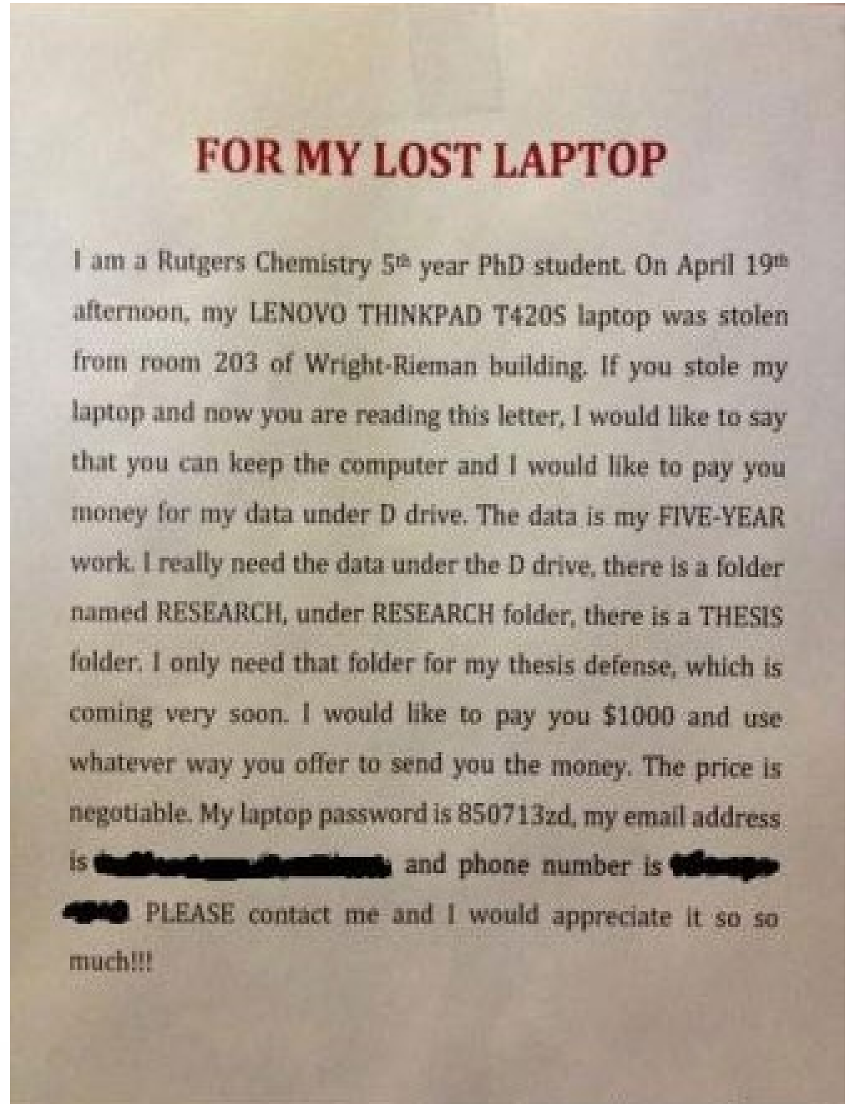
1. Backups

You need them

Situations to avoid



Lost backpack in MBTA orange line Chinatown station (to oak grove)

😕 Question

Dear all,

I am a Northeastern PhD student. I lost my blue backpack in Chinatown station (to oak grove) at ~7:30 pm today😭😭😭. I forgot to take it, which should be on one of the chairs on the right hand side when entering station. The backpack contains my thinkpad (contain my research work over 3 years😭😭😭), my HDD, my access to KRI, my phone charger. The backpack is super important to me!! If anyone picked it up or even has any clues, please leave me a message. Thanks a lot!

30 17 Share

"The backpack contains my thinkpad (contains my research work over 3 years), my HDD, my access to KRI.."

A note on USB drives

- USB drives are **transfer devices** – they are not intended for backup or storage.

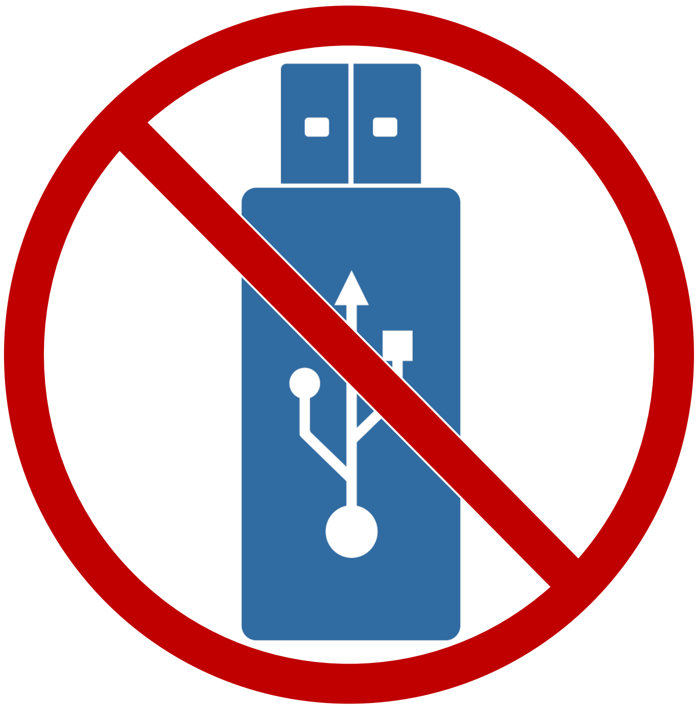


Photo credit: Amy Nurnberger, Storage fail, CC-By

The 3-2-1 backup strategy

- The **“3-2-1 rule”**, also known as **“the rule of 3”**:
  - **3 data copies:** Three copies of a file
  - **2 types of storage:** Two different media
  - **1 offsite location:** One copy geographically distant

https://eccezion.com/3-2-1-backup-rule/

Here + Near + Far: Applying the rule of 3

Here **local/working copy**

e.g., on your workstation or in shared workspace

Near **other local/external copy or in a remote location**

e.g., external hard drive (CDs / DVDs not built to last)

Far **remote copy**

e.g., on MIT's Code42 (CrashPlan) service or via private companies in the cloud

Test file recovery at setup and on a regular schedule

This is not for publishing or sharing (storage ≠ archiving)

Here + Near + Far = 3 copies (example 1)

https://commons.wikimedia.org/wiki/File%3ADesktop-PC.svg

https://pixabay.com/p-860234/ CC0

CRASHPLAN

Here

lab computer

Near

portable hard drive

(stored in fireproof box offsite)

Far

MIT's CrashPlan (Code42)

(on lab computer)

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Here + Near + Far = 3 copies (example 2)

https://pixabay.com/p-310098/ CC0

Here	Near	Far
laptop	**Dropbox**	**MIT's CrashPlan (Code 42)**
	(local files sync'd to Dropbox for business)	(on laptop)

CrashPlan (Code42)

Resources:

- Set Up CrashPlan
- IS&T Knowledgebase CrashPlan FAQ

To set up CrashPlan you need a Kerberos username & password.

What about lab/department computers that have multiple users?

- IS&T Help Desk can help set up a Code42 service account or a server backup as appropriate
- IS&T Contact

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Dropbox

Resources:

- Dropbox at MIT
- IS&T Knowledgebase page, Dropbox

Dropbox considerations:

- IS&T KB
    - "What sort of things should I not store in Dropbox?"
    - "What happens to my Dropbox for Business account when I leave MIT?"
- Dropbox Quota FAQ

OneDrive

Resources:

- OneDrive at MIT
- IS&T Knowledgebase page, OneDrive

OneDrive considerations:

- IS&T support for OneDrive seems minimal
- IS&T Knowledgebase, "What sort of things should I not store in the cloud?"
- OneDrive quota: 500GB
  OneDrive Quota Checking

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Additional data storage/backup resources

Resources:

- MIT Data Storage and Collaboration Options (includes individual & group/team enterprise solutions)
  Note: *This page includes a table comparing the different features available for storage options provided by MIT.*
- MIT Managed Servers
- Server backups with IBM Spectrum Protect (formerly TSM)
- Office of Research Compute and Data (ORCD)

2. File naming and organization

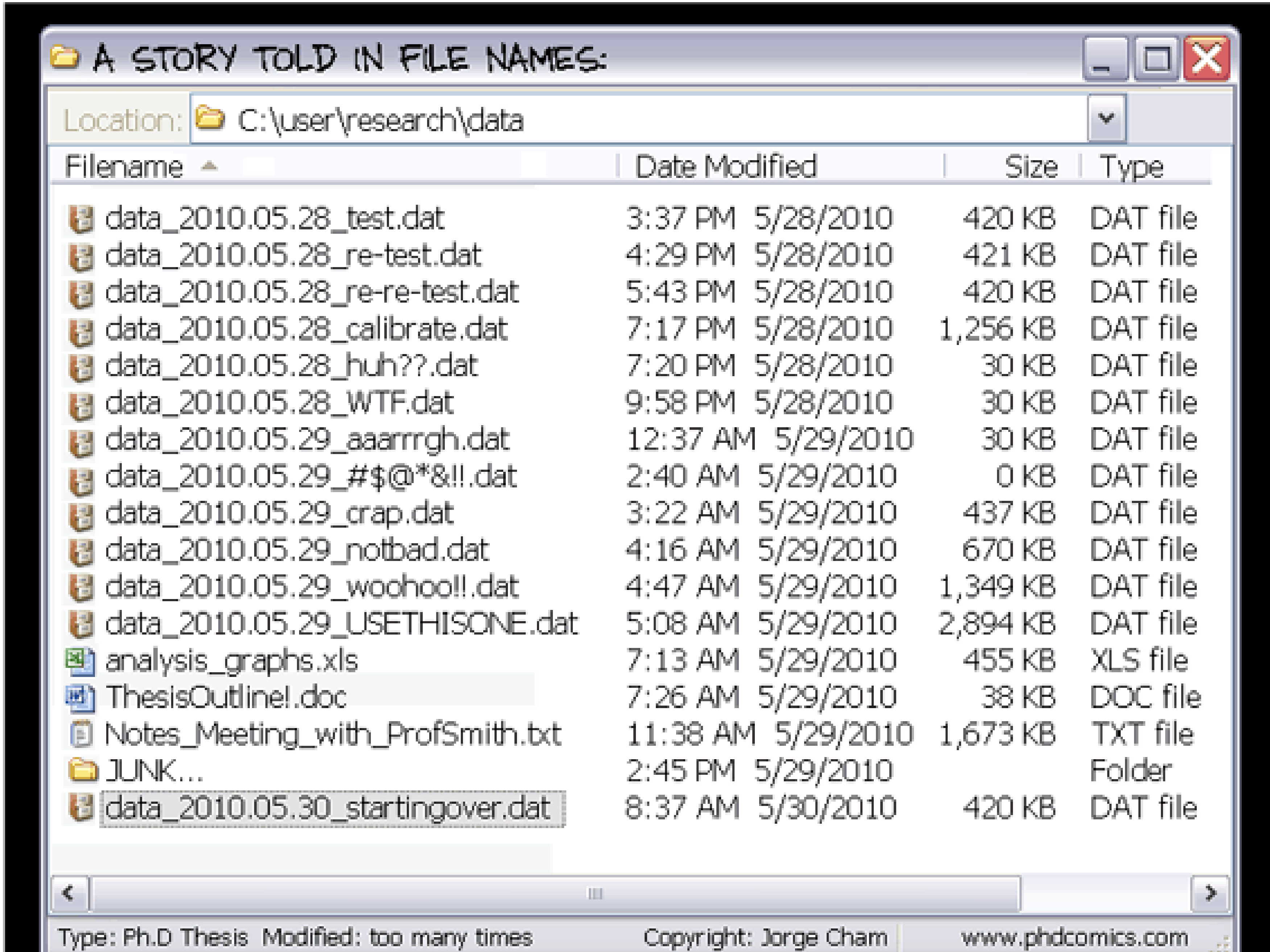
Dude, where's my stuff?

Morgan Edwards
@mangoedwards

I can't send you the original data because I don't remember what my excel file names mean anymore #overlyhonestmethods

12:11 PM - Jan 8, 2013

1 130 79



http://phdcomics.com/comics/archive.php?comicid=1323

File naming

Be descriptive. Be consistent.

- Use snake_case (with consistent letter casing) or camelCase.
- Don’t depend on letter casing to distinguish between files.
- Avoid whitespaces.
- Avoid overlapping names.
- Use normal characters and exclude special characters: & ,* % # ;*)(!@$^~' { [?<-.
- Use a period only <u>before</u> the file extension.
- Use numeric versioning (avoid using “final” or “copy” to distinguish file versions): <project_id>_v001, <project_id>_v002
- Use leading zeros when using sequential numbering.
- Use [ISO 8601 format](#) for dates: 20240101, 2024_01_01
- Limit file names to 32 characters or less.

See our [handouts & worksheets](#).

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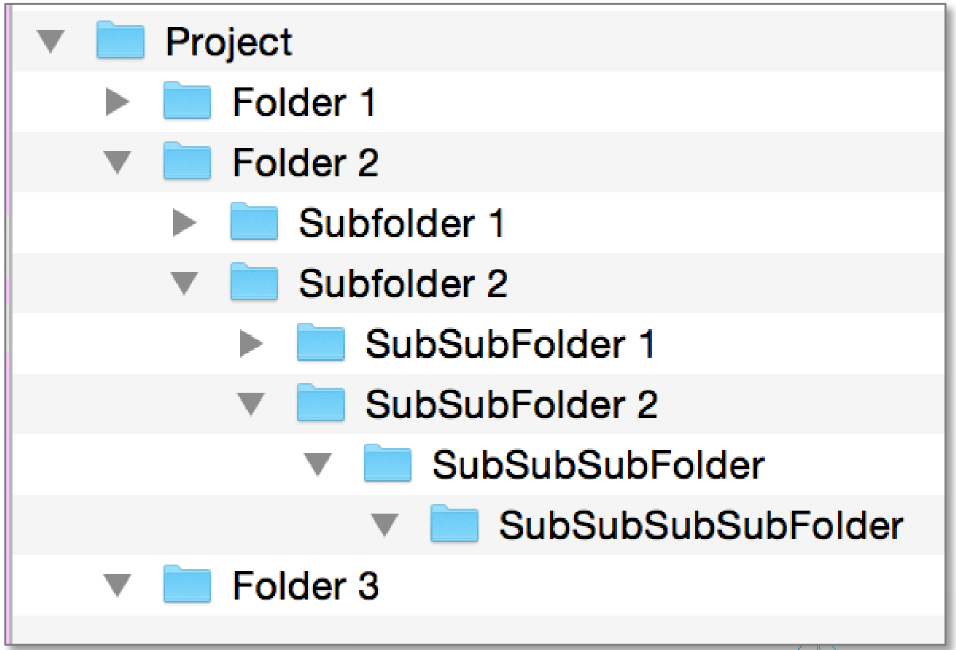
File structures

- Determine important contextual information (e.g., project name)
    - *When you* are looking for a file, how do you think about it?*
    - Avoid overlapping categories
- Don't let your folders get too big
- Don't let your structure get too deep



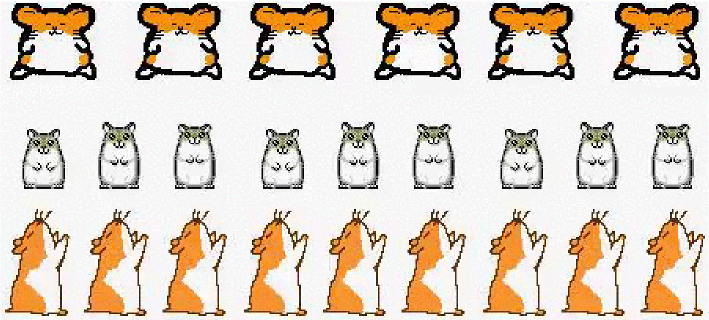
How many clicks does it take to get there?

File naming & organization mantra

Make **a system.** ***Share*** **the system.** ***Follow*** **the system.**

Be consistent.
Be consistent.
Be consistent.

Document.
Document.
Document.



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Additional file naming resources

Relevant workshop:

- Data Management: File Organization

Handouts:

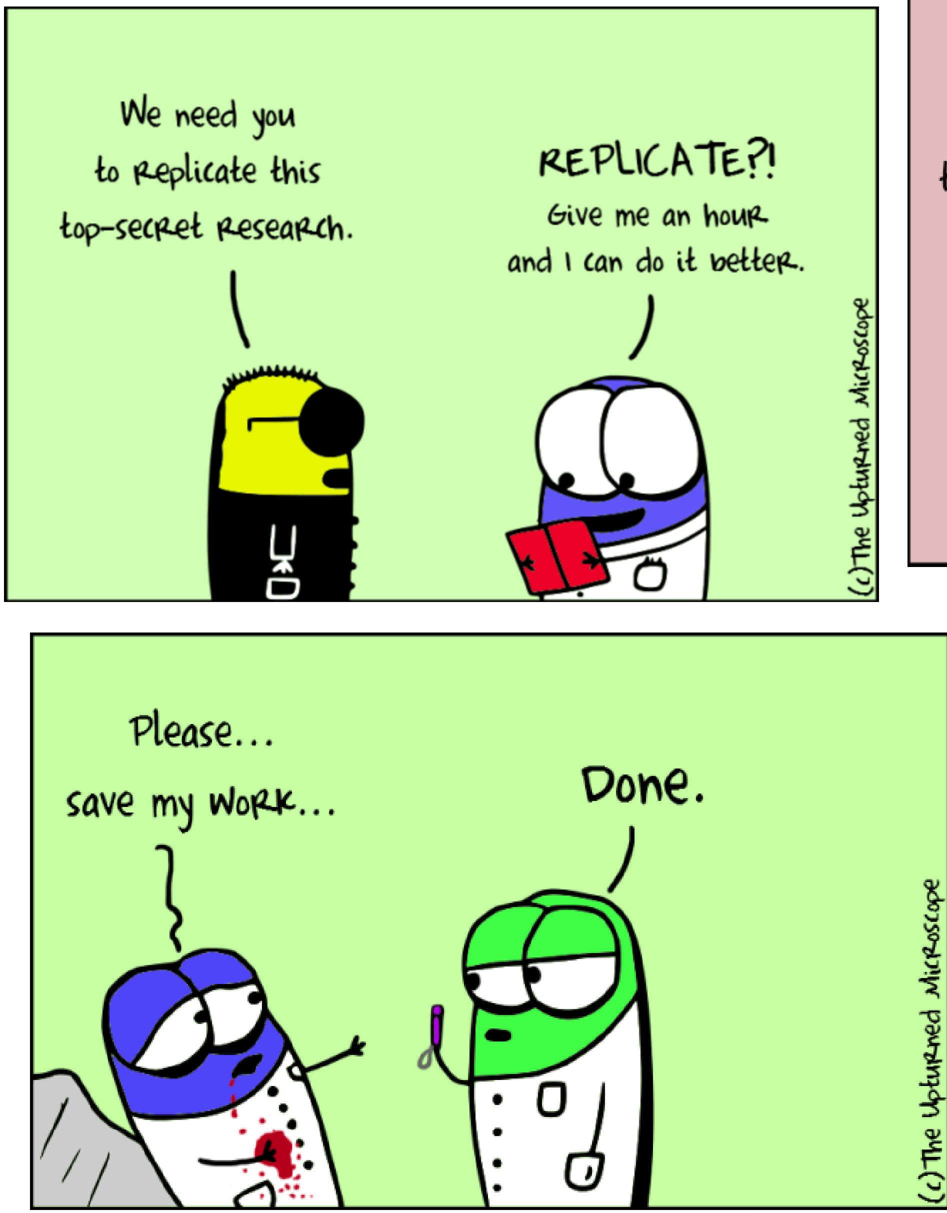
- Naming and organizing your files and folders worksheet
- Documenting your naming and organization schema, sample and template READMEs
- File naming best practices handout (pdf)
- Batch file renaming tools handout (pdf)
- Version control tools & techniques handout (pdf)

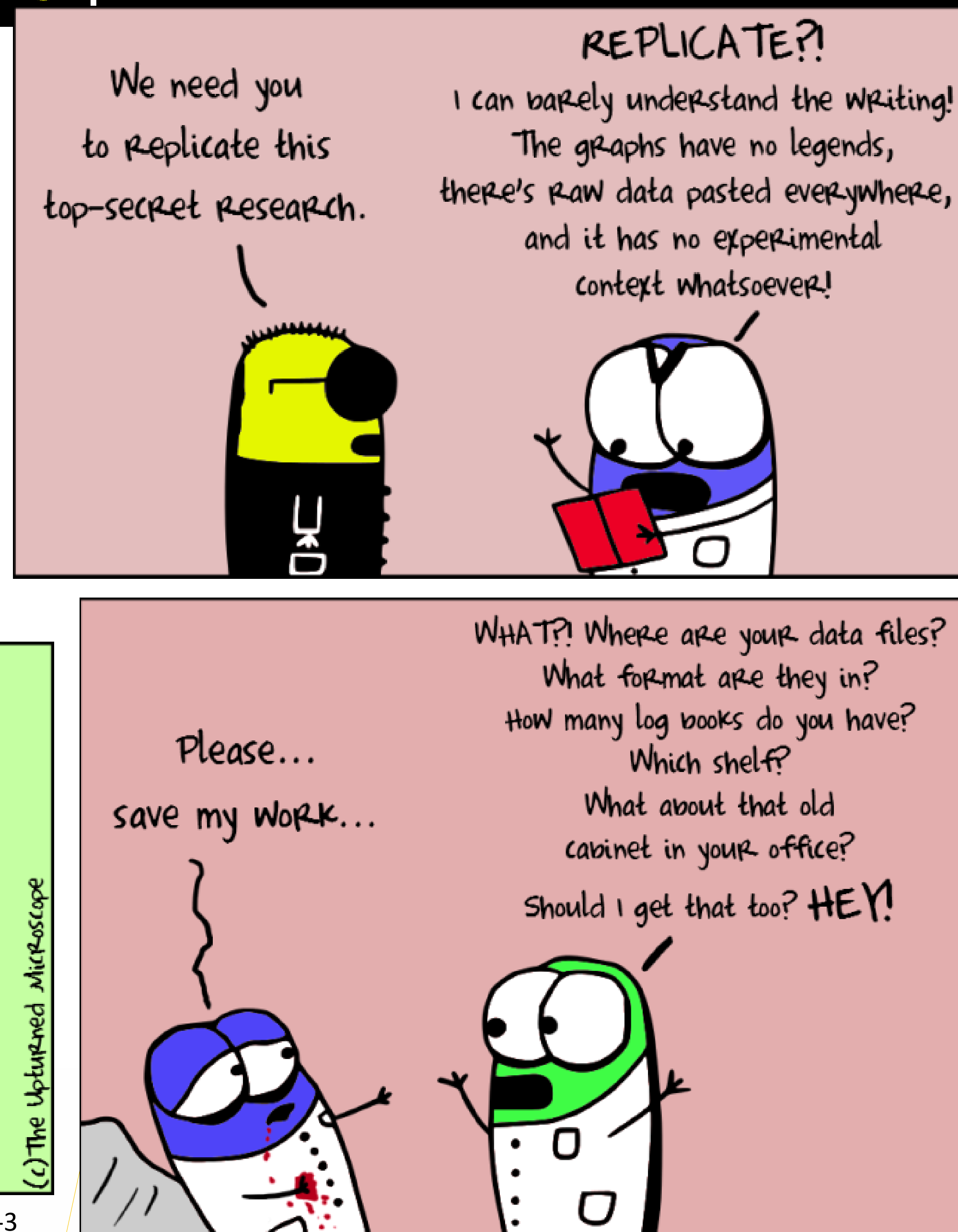
Also available on our website:
https://libraries.mit.edu/data-management/store/organize/

3. Documenting

Documentation is key.

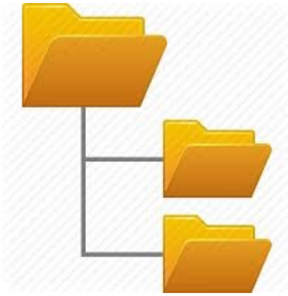
Movie scientist | Real scientist





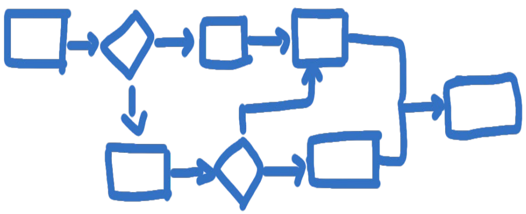
http://theupturnedmicroscope.com/comic/real-vs-movie-scientist-2/ & -3

What to include in documentation



Your organizational system

What are your file naming conventions? What is your folder hierarchy?



Your workflow

How did you get from raw data to the final product of research?

Your data

What is that data field? What are the units? What does that acronym mean?

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What is required to make use of your data

- Where to find it
- How to access it
- Intended uses of the data
- Known problems, inconsistencies, limitations
- Collection methods, units of measure, variable names
- Fixity checks
- Ethical/privacy restrictions
- Licensing
- Who to cite

Put it in a README file.

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Where documentation resides

labarchives
Chance Favors the Organized Lab

Lab notebooks

Lab / research group knowledge management

Open Science Framework

TXT

Plain text readme files

LabArchives (ELN)

LabArchives is a cloud-based electronic lab notebook (ELN). MIT faculty, staff, students, and affiliates have access to an enterprise license through IS&T.

Resources:

- Info on ELNs: https://libraries.mit.edu/data-management/store/electronic-lab-notebooks/
- To get started with LabArchives: go to http://labarchives.mit.edu, authenticate via Touchstone, and then activate an enterprise account.
- IS&T Knowledgebase page, LabArchives

Open Science Framework (OSF)

OSF is a cloud-based management system.

- Project structuring with wiki & dashboard
- File versioning
- Access control & collaboration capabilities
- Integrations with tools like Dropbox & GitHub

Resources:

- OSF for MIT: https://libraries.mit.edu/data-management/store/osf/
- Getting Started with OSF at MIT (1-page quick start PDF)
- MIT's OSF

README files

Best practices:

- Create one (1) README file for each data file/dataset
- Name the README so that it's easily associated with the data file(s) it describes
- Write your README document as a plain text file
- Format multiple README files identically (Tip: create & use a template)
- Follow the conventions for your discipline

Resources:

- Our Data bites: Writing better READMEs workshop
- Cornell's Guide to writing "readme" style metadata
- Template for Secondary data README

README files (example 1)

- The following is an example of a README file stored in a repository on Harvard Dataverse:
  - README from Estimating the Circulation and Climate of the Ocean (ECCO) Dataverse

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README example

- Comments:

- Each dataset listed above contains its own README file that provides more specific and detailed documentation.
- 'ECCO Version 4 Release 2' is a global ocean state estimate that covers the period from 1992 to 2011 (Forget et al. 2015, 2016). It was produced on behalf of the ECCO consortium (http://ecco-group.org/) with major support provided NASA's Physical Oceanography Program. General documentation of the 'ECCO Version 4 Release 2' dataverse and all included datasets can be found at https://dx.doi.org/10.7910/DVN/ODM2IQ (see README.pdf in that dataset).
- The formatting, online publishing, and archiving of the ECCO V4 R2 dataverse and datasets have benefited from guidance that was graciously provided by the MIT Libraries Data Management Services (http://libraries.mit.edu/data-management/). At time of writing the contents listed above can alternatively be downloaded from ftp://mit.ecco-group.org/ecco_for_las/version_4/release2/.

- References:

- Forget, G., J.-M. Campin, P. Heimbach, C. N. Hill, R. M. Ponte, and C. Wunsch, 2015: ECCO version 4: an integrated framework for non-linear inverse modeling and global ocean state estimation. Geoscientific Model Development, 8, 3071-3104, http://dx.doi.org/10.5194/gmd-8-3071-2015
- Forget, G., J.-M. Campin, P. Heimbach, C. N. Hill, R. M. Ponte, and C. Wunsch, 2016: ECCO Version 4: Second Release, http://hdl.handle.net/1721.1/102062

- Software:

- The ECCO V4 R2 files were produced using the `checkpoint64u' versions of the general circulation model (MITgcm and ECCO v4 settings) and Matlab analysis toolboxes (gcmfaces and MITprof). These software versions are available at http://mitgcm.org/download/other_checkpoints/ and http://mit.ecco-group.org/opendap/ecco_for_las/version_4/checkpoints/contents.html
- The up to date software documentations are available at http://mitgcm.org/public/r2_manual/latest/online_documents/manual.pdf, http://mitgcm.org/viewvc/*checkout*/MITgcm/MITgcm_contrib/gael/verification/eccov4.pdf, and http://mitgcm.org/viewvc/*checkout*/MITgcm/MITgcm_contrib/gael/matlab_class/gcmfaces.pdf

- Contact Us:

- questions regarding the ECCO model set-up, grid, software, or files should be addressed to either ecco-support@mit.edu (please subscribe via http://mailman.mit.edu/mailman/listinfo/ecco-support) or mitgcm-support@mitgcm.org more generally (please subscribe via http://mitgcm.org/mailman/listinfo/mitgcm-support).

- README file revision history:

- add c66a_eccov4r2.tar frozen copy of model setup [Gael Forget] [2016/12/07]
- README file overhaul for use within dataverse [Gael Forget] [2016/08/03]

https://doi.org/10.7910/DVN/PICCRE

README files (example 2)

- The following is an example of a README stored in a GitHub repository:
  - README from *The Pudding*, a digital publication, describing the data for one of their essays.

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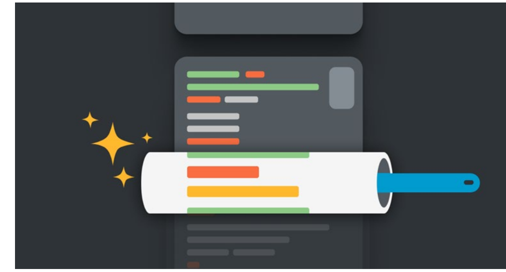
Documenting code



Code documentation helps users understand purpose and functionality of code, which is essential for future development.

- Add a README to your code repository
- Add comments and docstrings (“documentation strings”) throughout your code
  - Describe (a) what the function does, (b) inputs, and (c) return values.
  - Google’s style guide for comments and docstrings
  - Visual Studio Code autodocstring extension
- Use a version control system (e.g., Git) with a compatible developer platform (e.g., GitHub or GitLab)
- Use “linters” to identify semantic and stylistic errors in your code
  - Visual Studio Code guide to linting Python

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4. Security

Who can access your stuff?
(Who should be able to access your stuff?)

Security checkpoints

- Personally Identifiable Information
- Protected populations (human/non)
- Controlled Unclassified Information
- Data Use Agreements
- Human participants
- Proprietary data
- Patents

Image: https://pixabay.com/en/slippery-wet-caution-warning-151881/ CC0

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Security options

- Passwords & Password managers (LastPass, 1Password, etc.)
- Encryption
- Airgap
- Don't email the important things!
- De-identification & anonymization

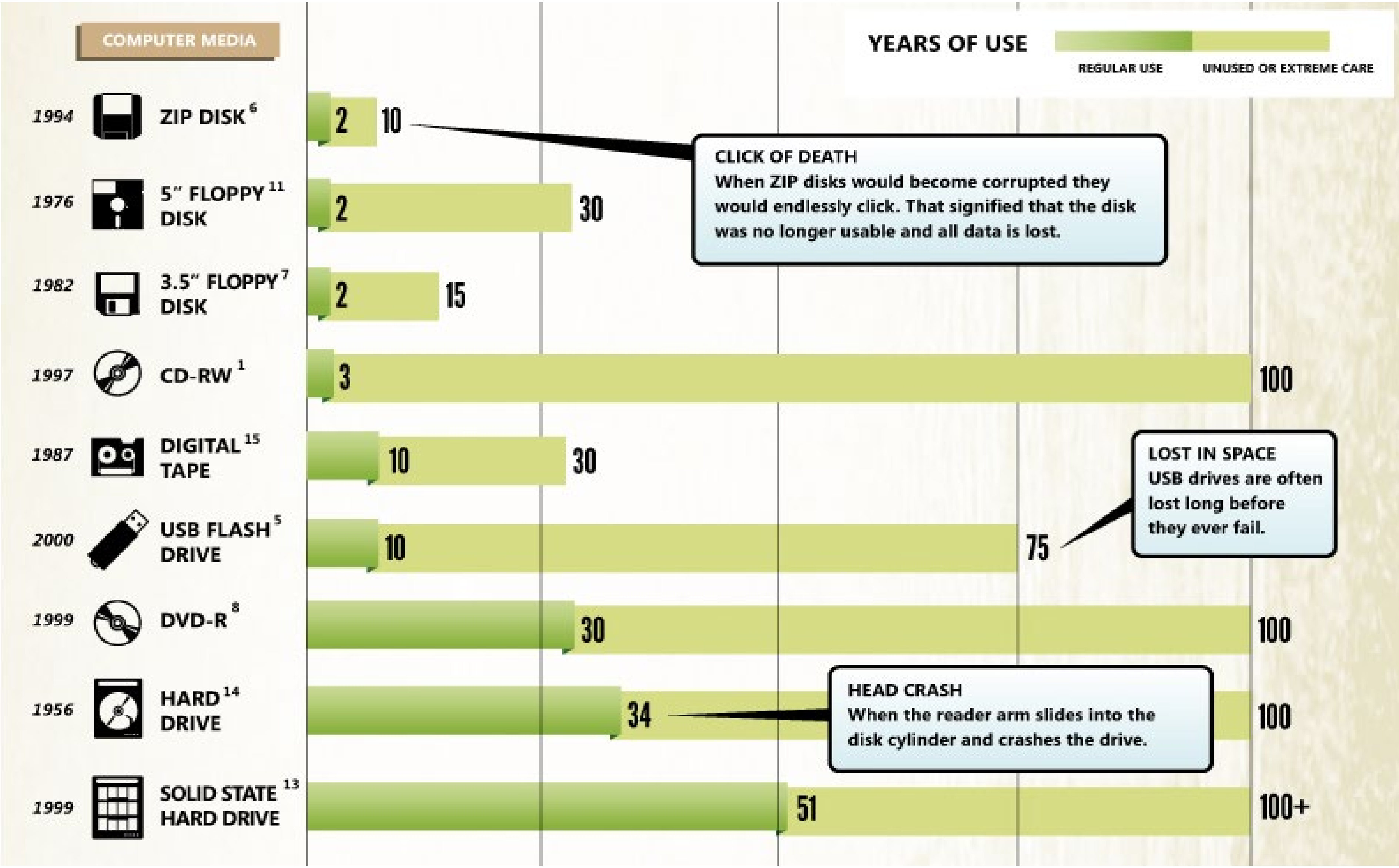
Resources:

- MIT COUHES, Guidelines for Selected Procedures and Populations
- MIT OSATT: DUAs & NDAs
- MIT's Written Information Security Program (WISP)
- MIT IS&T: Secure Computing & InfoProtect
- MIT VPR, Compliance: Security, Integrity, and Compliance

5. Future proofing

Store and share your data as openly as possible.

Strive for reproducibility.



Lifespan of Storage Media: http://www.crashplan.com/medialifespan/

File formats

In the best case, your data file formats are:

- Non-proprietary (also known as *open*)
- Unencrypted and uncompressed*
- Lossless*

File formats: Proprietary vs. Preferred

Proprietary Format	Alternative/Preferred Format
Excel (.xls, .xlsx)	Comma separated values (.csv, .tsv); ASCII
Word (.doc, .docx)	Plain text (.txt); PDF/A (.pdf)
PowerPoint (.ppt, .pptx)	PDF/A (.pdf)
Photoshop (.psd)	TIFF (.tif, .tiff)
Quicktime (.mov)	MPEG-4 (.mp4)

Not sure about a file format?
Check https://www.nationalarchives.gov.uk/PRONOM/default.htm

Library of Congress Recommended Format Specifications (for Preservation)
http://www.loc.gov/preservation/resources/rfs/

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File formats: loss

Converting file formats? Be aware of information loss!

To mitigate the risk of lost information when converting:

- Note the conversion steps you take
- If possible, keep the original file as well as the converted ones

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Storing your data for the long term

Use a Research Data Repository!

- Repositories offer storage and access for the longer term
- You don't have to support data requests!
- You get credit with a unique identifier to cite the data
- Can be discipline focused or general
- Desirable repository characteristics:
  - NSTC, Desirable Characteristics of Data Repositories for Federally Funded Research
  - NIH, Selecting a Repository for Data Resulting from NIH-Supported Research
  - The TRUST Principles for digital repositories

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

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Find a data repository

Look here!

- Zenodo – https://zenodo.org (connects to GitHub)
- Harvard Dataverse - https://dataverse.harvard.edu/
- Dryad - https://datadryad.org/stash (we are members!)

Specialized:

- ICPSR (social sciences)
- Qualitative Data Repository (QDR)
- See a directory at https://re3data.org or ask Data Management Services for help!

Questions? Comments? Tips?

Check out our web site: http://libraries.mit.edu/data-management

Upcoming workshops:
https://libraries.mit.edu/news/category/data/

Contact: data-management@mit.edu

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